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# Database Foundations

**1-3**

## **Types of Database Models**

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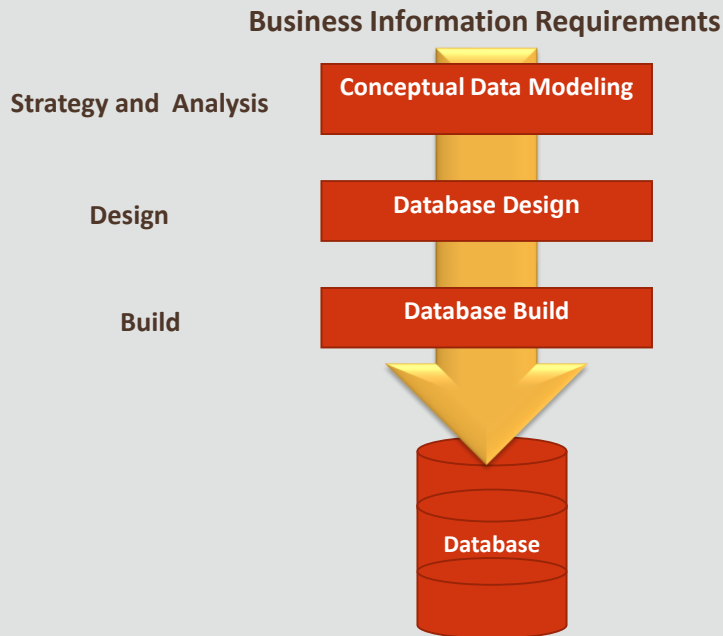
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# Objectives

- This lesson covers the following objectives:
  - Describe the database development process
  - Explain the common types of database models:
    - Flat file model
    - Hierarchical model
    - Network model
    - Object-oriented model
    - Relational model

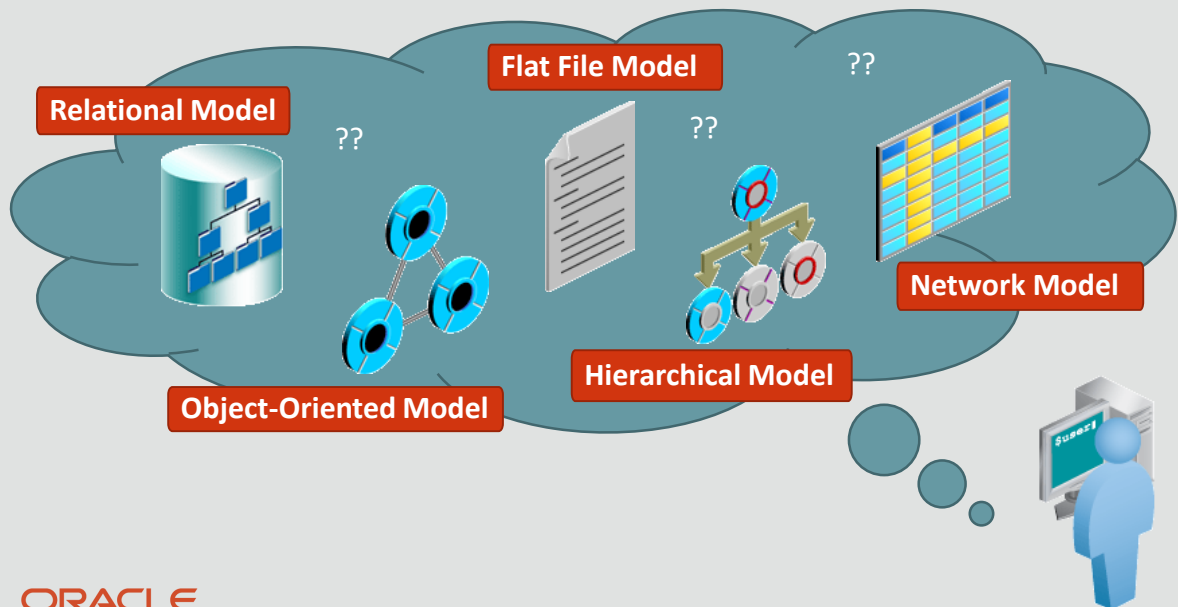


# Database Development Process

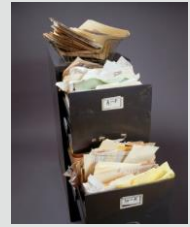


Data modeling is the first part of the database development process. Conceptual data modeling is the examination of a business and business data to determine the structure of business information and the rules that govern it. This structure forms the basis for database design. A conceptual model is relatively stable over long periods of time. Physical data modeling (or database building) is concerned with implementation in a given technical software and hardware environment. The physical implementation is highly dependent on the current state of technology and is subject to change as available technologies rapidly change.

# Case Scenario: Types of Database Models



# Flat File Model

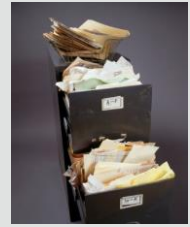


- A flat file database is a database designed around a single table
- Flat file databases are generally in plain-text form, where each line holds only one record.
- The fields in the record are separated with delimiters, such as tabs and commas.

Flat files serve as a solution for simple database tasks. The flat file design puts all database information in one table, or list, with fields to represent all parameters. A flat file may contain many fields, often with duplicated data that is prone to data corruption.

## Example of a Flat File Model

- Books as well as Authors are stored in this single table, causing repetition of data values



|          | AUTHOR_ID | AUTHOR_NAME      | TITLE                |
|----------|-----------|------------------|----------------------|
| Record 1 | AD0001    | Oscar Wilde      | A Vision             |
| Record 2 | AD0002    | Leo Tolstoy      | War and Peace        |
| Record 3 | AD0003    | Oliver Goldsmith | Citizen of the World |
| Record 4 | AD0003    | Oliver Goldsmith | The Deserted Village |

# Hierarchical Model

- In a hierarchical database model, the data is organized in a tree-like structure
- The data is stored as records that are connected to one another through links.
- A record is a collection of fields
- A record in the hierarchical database model corresponds to a row in the relational database model

Each field contains only one value. The entity type of a record defines which fields the record contains.

A record in the hierarchical database model corresponds to a row in the relational database model. An entity type corresponds to a table.

In a hierarchical database model:

- Each child record has only one parent.
- A parent record can have one or more child records.

To retrieve data from a hierarchical database, the whole tree needs to be traversed, starting from the root node.

Advantages:

- Easy addition and deletion of new information
- Faster access to data at the top of the hierarchy

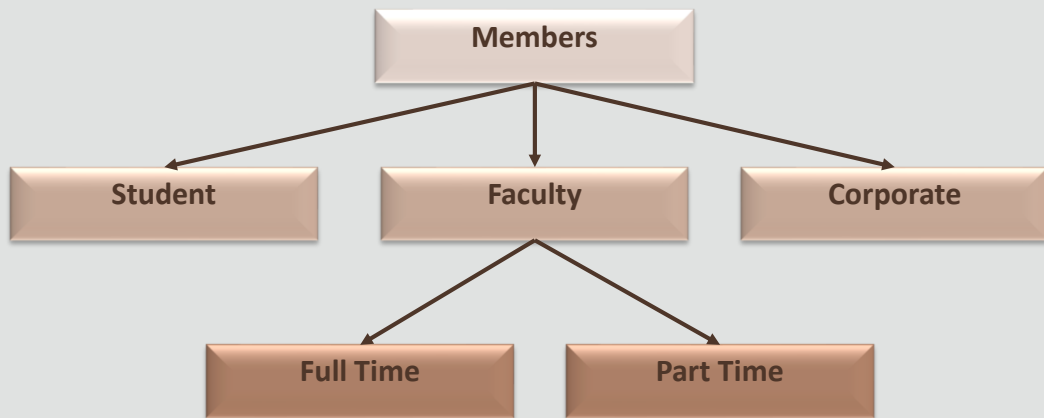
Disadvantages:

- Increased storage space
- Slower access to data at the bottom of the hierarchy



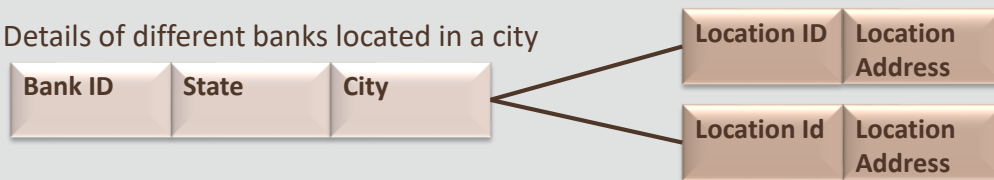
## Example of a Hierarchical Model

- Data is organized in a tree-like structure and stored as records that are connected to one another through links



# Network Model

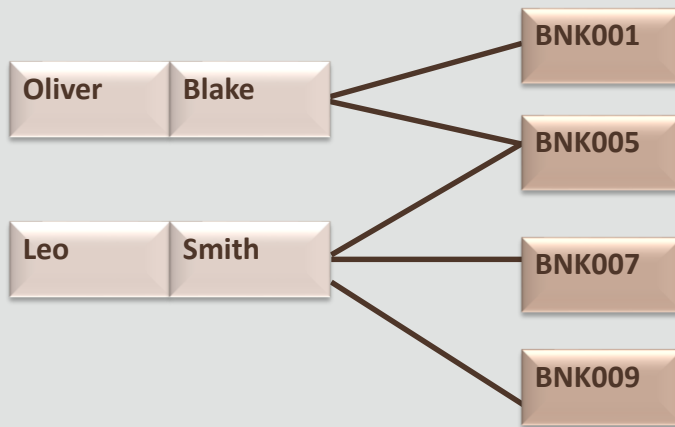
Details of different banks located in a city



- The network model is a database model that can be regarded as a flexible way of representing objects and their relationships
- A network database comprised of a collection of records connected to one another through links. (Boxes = Fields, Lines = Links)
- Each record is a collection of fields, each of which contains only one data value
- A link is an association between two records

In a network database model each record can have multiple parent and child records, forming a generalized graph structure. The network model enables a more natural way of modeling the relationship between records.

## Example of a Network Model



- Oliver Blake holds accounts in two banks, BNK001 and BNK005
- Leo Smith holds accounts in three banks, BNK005, BNK007, BNK009

The slide depicts an example of a network model that stores information about bank account details of different people. In the example, The records are connected to each other through links, represented by lines.

# Object-Oriented Model

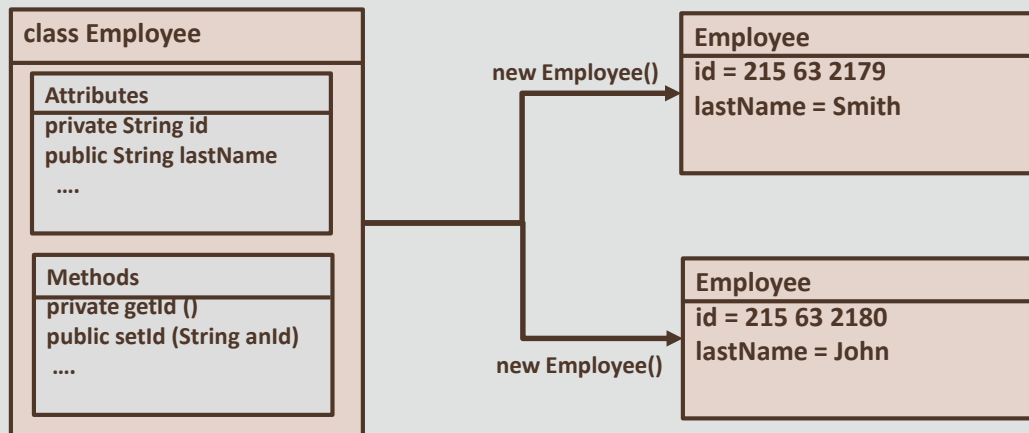
- A entity is modeled as an object
- Every object has a state (the set of values for the attributes of the object) and a behavior (the set of methods that operate on the state of the object)
- The relationship between the objects is through sharing of access
- An object must belong to only one class as an instance of that class
- You can derive a new class (subclass) from an existing class (superclass)

## Advantages

- Reduced maintenance
- Real-world modeling
- High code reusability

Disadvantages: Many information application systems do not benefit from object-oriented modeling because it is best suited for dynamic, interactive environments.

# Example of an Object-Oriented Model



- Shown are two Employee objects created from the Employee class
- Each with different values for the attributes of id and lastName

The graphic in the slide shows an Employee class defined with two attributes:

- The id attribute is the employee identifier.
- The lastName attribute is the last name of the employee.

The Employee class has two methods:

- getId()
- setId(String anId)

The id attribute and the getId() method are private and therefore can be accessed only within the class. The lastName attribute and the setId(String anId) method are public and can be accessed by other classes, too.

When you create an instance, the attributes store individual and private information relevant only to the employee. The information contained within an employee instance is known only to that particular employee. Each instance of Employee holds its own state. You can access that state only if the creator of the class defines it in a way that gives you access.

# Relational Model

- Data is represented as a collection of tables
- Each column represents attributes that belong to the table
- Each row represents an instance of the table
- Each table is the visual representation of columns and rows
- Every table has a field or a set of fields that uniquely identifies the row

# Relational Model

- The order of the rows and columns is not important
- Every row is unique
- Each field can contain only one value
- Values within a column or field are from the same domain (datatype)
- Table names must be unique
- Column names within each table must be unique

# Example of a Relational Model

EMPLOYEE

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | DEPARTMENT_ID |
|-------------|------------|-----------|---------------|
| 100         | Steven     | King      | 90            |
| 101         | Neena      | Kochhar   | 90            |
| 102         | Lex        | De Haan   | 90            |
| 200         | Jennifer   | Whalen    | 10            |
| 205         | Shelley    | Higgins   | 110           |

Foreign Key

In this example a relationship is created between the two tables using the common field of DEPARTMENT\_ID

DEPARTMENT

| DEPARTMENT_ID | DEPARTMENT_NAME |
|---------------|-----------------|
| 10            | Administration  |
| 20            | Marketing       |
| 50            | Shipping        |

refers to

Primary Key

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Types of Database Models

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In the slide example, the Employees table contains a column that relates to the Department\_ID from the Departments table. The inclusion of the Department\_ID defines the relationship in the relational database model.



# Summary

- In this lesson, you should have learned how to:
  - Describe the database development process
  - Explain the common types of database models:
    - Flat file model
    - Hierarchical model
    - Network model
    - Object-oriented model
    - Relational model



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